



NCCER Module 22107



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Vertical-Mast Sit-Down Counterbalance Forklifts





Section Five

Basic Preventive Maintenance

Objective

5. Explain the difference between scheduled maintenance and daily inspections.
 - a. Describe the difference between required and scheduled maintenance and what inspections are performed at each of the major intervals.
 - b. Describe the types of inspections that should occur during daily forklift inspection.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

4. Perform a pre-start inspection and maintenance check.



5.0.0 Basic Preventive Maintenance

Scheduled forklift maintenance is essential. An operator should always be aware of the forklift's total hours of service and have the machine checked when the hours equal the pre-determined time interval.

Benefits of regular maintenance include:

- Reduced downtime
- Reduced equipment costs
- Improved safety
- Higher productivity
- OSHA compliance



5.1.0 Scheduled Inspections

Maintenance time intervals established by SAE are:

- 10 hours (daily)
- 50 hours (weekly)
- 250 hours (3 months)
- 500 hours (6 months)
- 1,000 hours (1 year)
- 2,000 hours (2 years)
- 3,000 hours (3 years)



5.2.0 Pre-Operation and Daily Inspections (1 of 5)

- *OSHA 29 CFR 1910.178* guideline mandates that forklifts be inspected daily and, when an issue is discovered, it must be reported, and the forklift removed from service until repaired by authorized personnel.
- **Walk-arounds** include both visual and operational checks of the forklift based on recommendations included in the operator's maintenance manual.
- A daily inspection checklist can be obtained from your employer, the manufacturer's operating manual, or from OSHA.
- Some forklifts may have a decal on them that includes complete daily and weekly inspection checklists.



5.2.0 Pre-Operation and Daily Inspections (2 of 5)

Daily Inspection for IC-Powered Forklifts:

- Engine oil and coolant levels
- Radiator and associated hoses
- Fuel level
- Hydraulic fluid level
- Accelerator and brake pedals connections and functions
- Parking brake operation
- If the forklift uses LPG, ensure the tank is mounted properly, that it fits the profile of the machine, and that there are no leaks in the hoses and connections.



5.2.0 Pre-Operation and Daily Inspections (3 of 5)

Daily Inspection for Electric-Powered Forklifts:

- Battery cables, wires, and mounts
- Hydraulic fluid level
- Battery fluid level



5.2.0 Pre-Operation and Daily Inspections (4 of 5)

Daily Inspection for All Conventional Forklifts:

- Check around and under the machine for leaks, damaged components, cracked welds, or missing bolts or pins.
- Inspect all attachments for wear and damage. Make sure there is no damage that would create unsafe operating conditions or cause an equipment breakdown.
- Inspect and clean steps, walkways, and handholds.
- Inspect the engine compartment and remove any debris.
- Inspect the hydraulic assembly for leaks, faulty hoses, or loose clamps.
- Inspect the lights and replace any broken bulbs or lenses.



5.2.0 Pre-Operation and Daily Inspections (5 of 5)

- Inspect axles, differentials, wheel brakes, and transmission for leaks.
- Inspect tires for damage and replace any missing valve caps. Ensure that the tires are inflated to the correct pressure. Many forklifts have hard-rubber tires; some have foam-filled tires; and others have non-pneumatic tires.
- Check the condition and adjustment of drive belts on the engine.
- Inspect the operator's cab and remove any trash.
- Inspect the windows for visibility and clean them, if necessary.
- Adjust the mirrors.
- Test the horn.



5.2.1 Servicing a Forklift

Follow the manufacturer's recommendations and service chart. Hydraulic fluids and filters should be changed during scheduled maintenance or whenever the fluid breaks down due to overheating.

Prior to performing maintenance, always do the following:

- Park the forklift on a level surface to ensure that fluid levels display correctly.
- Be sure the F-N-R lever is in Neutral.
- Lower the forks and attachments to the ground.
- Ensure the parking brake is set.
- Turn the engine start switch to the Off position on IC-powered forklifts.



5.2.2 Keeping Maintenance Records

- Accurate, up-to-date maintenance records are essential for knowing a forklift's history.
- Create a manual that lists the inspections and servicing activities required for each interval.
- An operator's manual and an inspection sheet should always be kept with the forklift.



Wrap Up – Trade Terms

Maintenance Time intervals: The periods of time between scheduled forklift inspections that are established by the Society of Automotive Engineers (SAE).

Preventive Maintenance: Scheduled service designed to prevent breakdowns and maintain peak equipment performance. This type of service is usually based on the forklift's total hours of operation.

Required Maintenance: Service that results from issues discovered during walk-around inspections or from a problem that interferes with the forklift's normal operation.

Walk-arounds: Daily visual and operational inspections that occur prior to operating the forklift.



5.0.0 Section Review Question 1

1. Which of the following are benefits of implementing a preventive maintenance program?
 - a. Higher productivity and reduced downtime
 - b. Longer lasting oil and transmission fluid
 - c. Longer lasting gauges and instruments
 - d. Assurance the lift will not break down



5.0.0 Section Review Question 1 Answer

1. Which of the following are benefits of implementing a preventive maintenance program?
 - a. Higher productivity and reduced downtime**
 - b. Longer lasting oil and transmission fluid
 - c. Longer lasting gauges and instruments
 - d. Assurance the lift will not break down



5.0.0 Section Review Question 2

2. Using the typical periodic maintenance schedule in Appendix A, which of the following should occur at 500 hours of service?
- a. Change the transmission oil and filter
 - b. Replace seat belts
 - c. Change engine oil and filter
 - d. Torque the engine head bolts



5.0.0 Section Review Question 2 Answer

2. Using the typical periodic maintenance schedule in Appendix A, which of the following should occur at 500 hours of service?
- a. Change the transmission oil and filter
 - b. Replace seat belts
 - c. Change engine oil and filter**
 - d. Torque the engine head bolts



5.0.0 Section Review Question 3

3. Walk-around inspections are performed _____.
a. At the end of each work week
b. When the operator notices a leak
c. When the operator is finished with the lift
d. Prior to operating the forklift



5.0.0 Section Review Question 3 Answer

3. Walk-around inspections are performed _____.
a. At the end of each work week.
b. When the operator notices a leak.
c. When the operator is finished with the lift.
d. Prior to operating the forklift.



5.0.0 Section Review Question 4

4. Which of the following is an example of scheduled maintenance?
- a. Changing the engine oil and filter
 - b. Checking the battery water level
 - c. Inspecting the oil and coolant levels
 - d. Adjusting the parking brake



5.0.0 Section Review Question 4 Answer

4. Which of the following is an example of scheduled maintenance?
- a. Changing the engine oil and filter**
 - b. Checking the battery water level
 - c. Inspecting the oil and coolant levels
 - d. Adjusting the parking brake



5.0.0 Section Review Question 5

5. Which of the following is an example of a daily inspection item?
- a. Changing the engine oil and filter
 - b. Adjusting the clutch
 - c. Inspecting the oil and coolant levels
 - d. Inspecting carriage rollers and chains



5.0.0 Section Review Question 5 Answer

5. Which of the following is an example of a daily inspection item?
- a. Changing the engine oil and filter
 - b. Adjusting the clutch
 - c. Inspecting the oil and coolant levels**
 - d. Inspecting carriage rollers and chains



Next...

Section 6.0.0

Forklift Operation



Read Sections 6.0.0 through 6.5.6 and complete the Section Review.